Haruto Sato

Robotics Software Engineer

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PROFESSIONAL SUMMARY

Built ROS nodes for autonomous inspection robots. Implemented OpenCV based defect detection and sensor fusion for camera and lidar data. Improved object localization accuracy 18 percent in warehouse tests.

TECHNICAL SKILLS

C++, ROS, Python, computer vision, OpenCV, Linux, sensor fusion

PROFESSIONAL EXPERIENCE

Built production REST APIs using Python and FastAPI.

Worked with PostgreSQL and Redis for backend data services.

Wrote automated tests and CI/CD pipelines for production releases.

Robotics Software Engineer | Vertex Systems | 2023 - Present

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- Owned technical planning, code reviews, and production support for work aligned to C++ and adjacent systems.

EDUCATION

M.S. Robotics, Carnegie Mellon University, 2022

SELECTED PROJECTS

Robotics Software Engineer portfolio project: built and documented a production-style solution using C++, with tests, monitoring, and a short implementation guide.

CERTIFICATIONS AND ADDITIONAL INFORMATION

Available for collaborative product development, technical communication, and structured problem solving. References available on request.